

Set 16: Customer Demographics Analysis

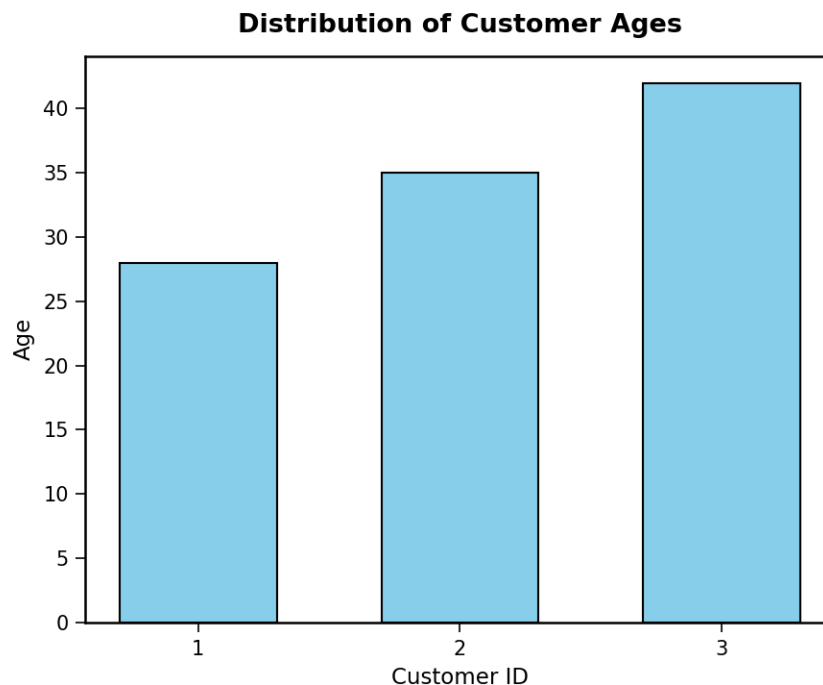
a) Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

R code:

```
# Dataset
customer <- data.frame(
  CustomerID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)
```

```
# Bar Chart
barplot(customer$Age,
  names.arg = customer$CustomerID,
  col = "skyblue",
  xlab = "Customer ID",
  ylab = "Age",
  main = "Distribution of Customer Ages")
```

output:



b) Generate a pie chart to represent the distribution of customers by gender.

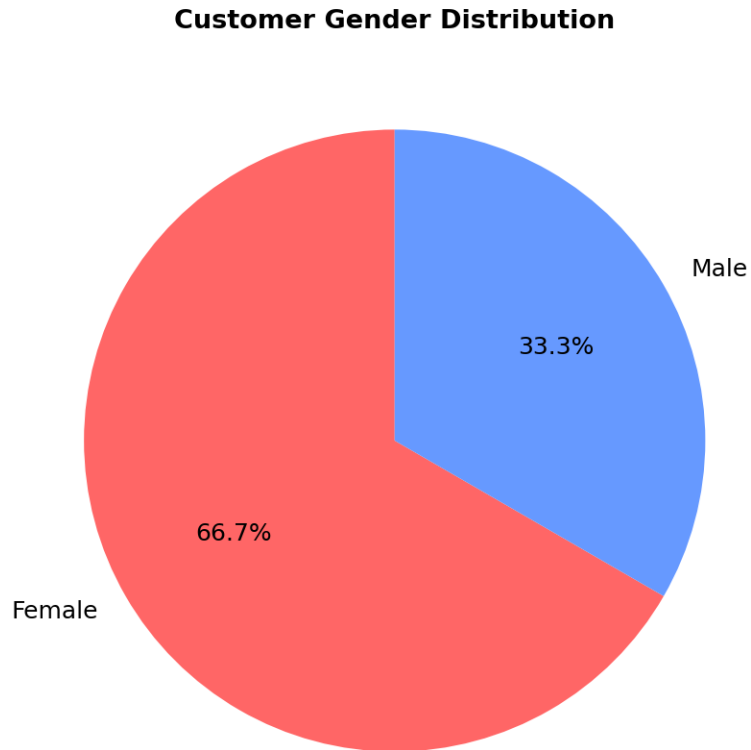
R code:

```
# Count Gender
gender_count <- table(customer$Gender)

# Pie Chart
pie(gender_count,
```

```
labels = paste(names(gender_count), round(100*gender_count/sum(gender_count),1),
"%"),
col = c("pink", "lightblue"),
main = "Customer Gender Distribution")
```

output:



c) Build a table to show the demographic information for each customer. Label the table elements.

R code:

```
# Display the table
print(customer)

# Formatted table
install.packages("gridExtra")
library(gridExtra)
grid.table(customer)
```

output:

CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

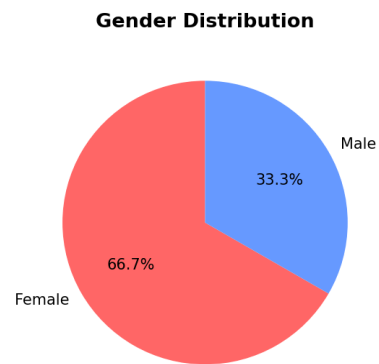
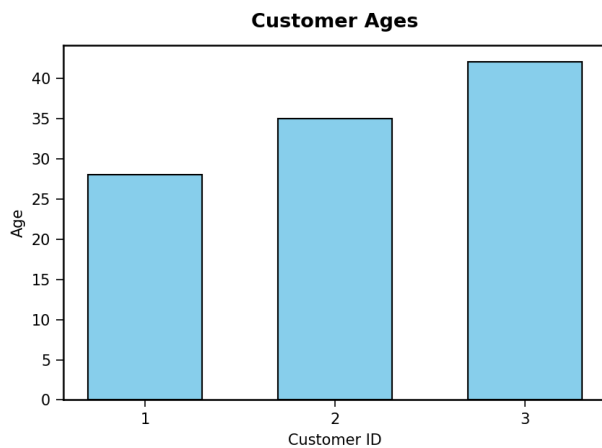
d) Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

R code (dashboard simulation):

```
par(mfrow = c(1, 2))
barplot(customer$Age,
  names.arg = customer$CustomerID,
  col = "skyblue",
  xlab = "Customer ID", ylab = "Age",
  main = "Customer Ages")
pie(gender_count,
  labels = names(gender_count),
  col = c("pink", "lightblue"),
  main = "Gender Distribution")
```

output:

Customer Demographics Dashboard



Set 17: Employee Performance Analysis

a) Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

R code:

```
# Dataset
employee <- data.frame(
  EmployeeID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  YearsService = c(5, 3, 7),
```

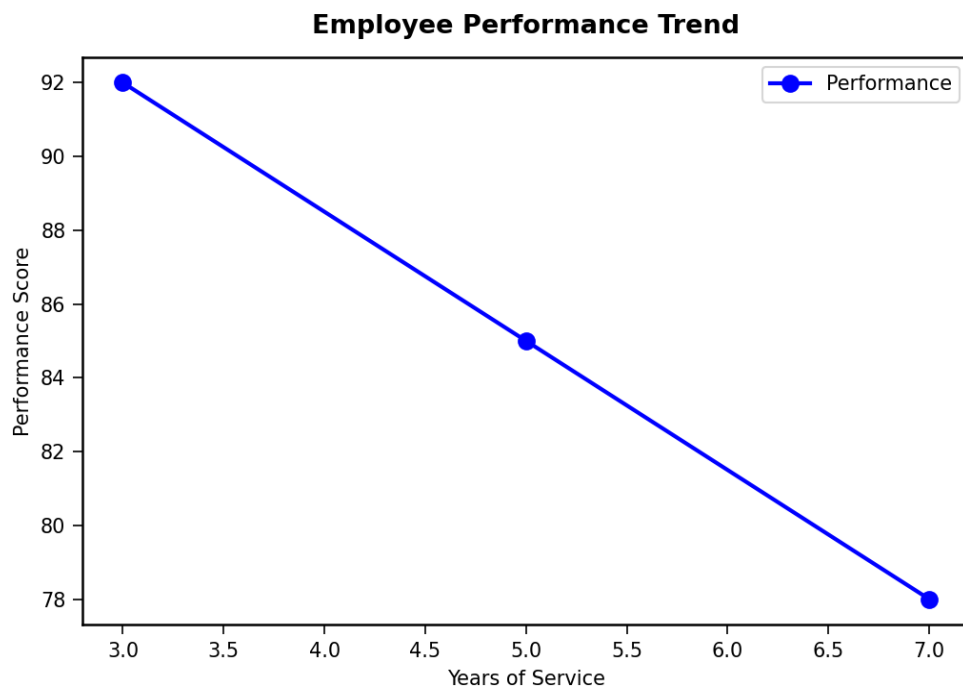
```

Performance = c(85, 92, 78)
)

# Line Chart
employee <- employee[order(employee$YearsService), ]
plot(employee$YearsService,
      employee$Performance,
      type = "o",
      col = "blue",
      pch = 16,
      lwd = 2,
      xlab = "Years of Service",
      ylab = "Performance Score",
      main = "Employee Performance Trend")
legend("topright", legend = "Performance", col = "blue", lty = 1, pch = 16)

```

output:



b) Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

R code:

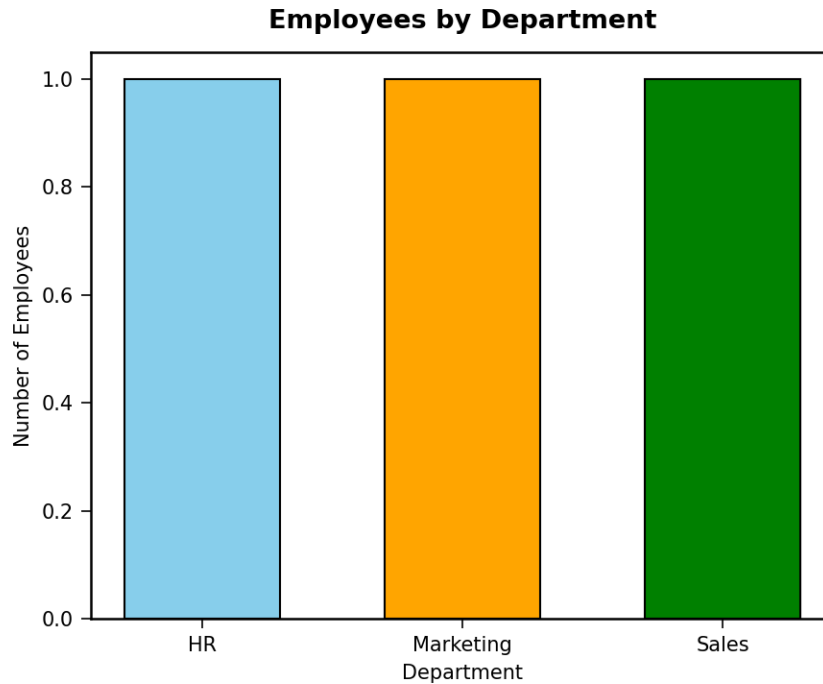
```

# Department count
dept_count <- table(employee$Department)

# Bar Chart
barplot(dept_count,
        col = c("skyblue", "orange", "green"),
        xlab = "Department",
        ylab = "Number of Employees",
        main = "Employees by Department")

```

output:



c) Build a table to display the performance data for each employee. Label the table elements.

R code:

```
print(employee)
install.packages("gridExtra")
library(gridExtra)
grid.table(employee)
```

output:

EmployeeID	Department	YearsService	Performance
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

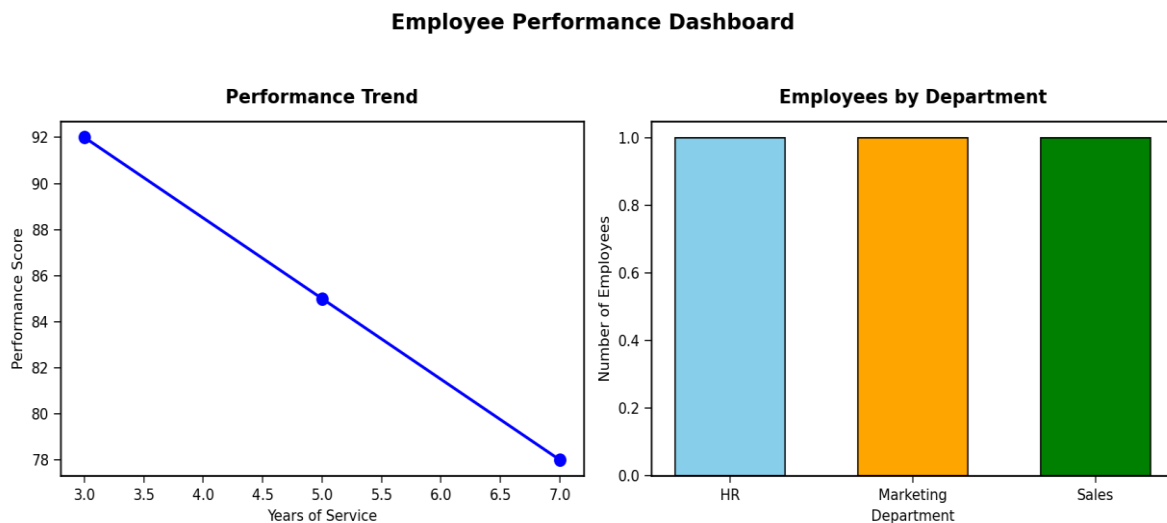
d) Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

R code (dashboard simulation):

```
par(mfrow = c(1, 2))
plot(employee$YearsService,
     employee$Performance,
     type = "o", col = "blue", pch = 16, lwd = 2,
     xlab = "Years of Service", ylab = "Performance Score",
     main = "Performance Trend")
barplot(dept_count,
     col = c("skyblue", "orange", "green"),
```

```
xlab = "Department", ylab = "Number of Employees",
main = "Employees by Department")
```

output:



Set 18: Product Inventory Management

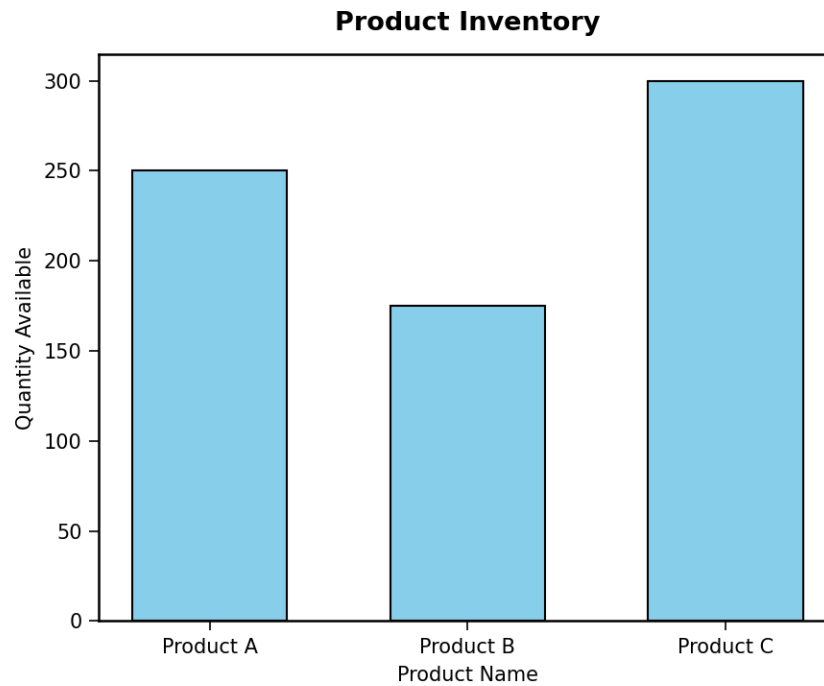
a) Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

R code:

```
# Dataset
inventory <- data.frame(
  ProductID = c(1, 2, 3),
  ProductName = c("Product A", "Product B", "Product C"),
  Quantity = c(250, 175, 300),
  Price = c(20, 15, 18)
)
```

```
# Bar Chart
barplot(inventory$Quantity,
  names.arg = inventory$ProductName,
  col = "skyblue",
  xlab = "Product Name",
  ylab = "Quantity Available",
  main = "Product Inventory")
```

output:



b) Generate a stacked bar chart to show the quantity of each product within different product categories.

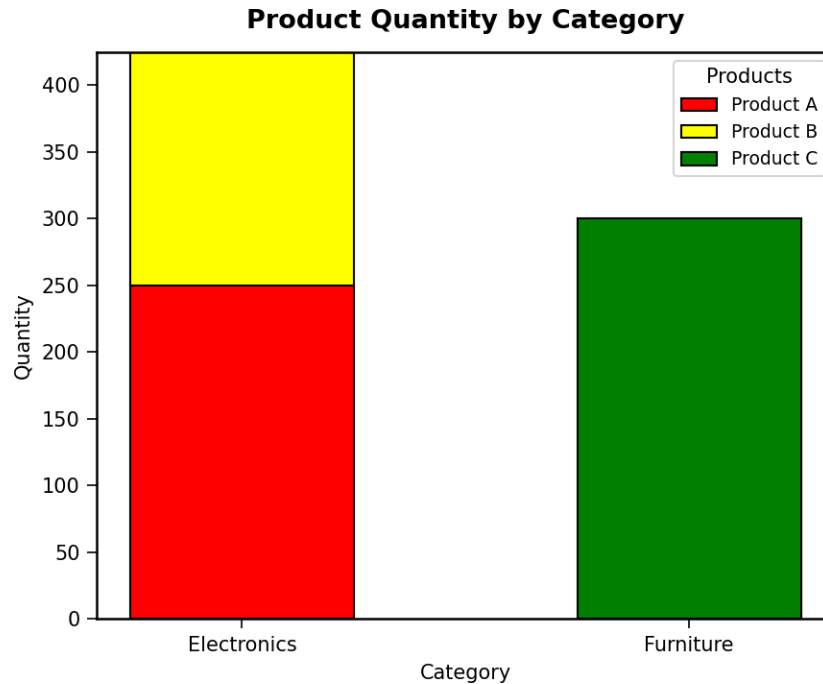
R code:

```
# Add Category
inventory$Category <- c("Electronics", "Electronics", "Furniture")

# Frequency Table
table_data <- xtabs(Quantity ~ Category + ProductName, data = inventory)

# Stacked Bar Chart
barplot(table_data,
  beside = FALSE,
  col = rainbow(ncol(table_data)),
  main = "Product Quantity by Category",
  xlab = "Category",
  ylab = "Quantity")
legend("topright", legend = colnames(table_data), fill = rainbow(ncol(table_data)),
title = "Products")
```

output:



c) Build a table to show the inventory data for each product. Label the table elements.

R code:

```
print(inventory)
install.packages("gridExtra")
library(gridExtra)
grid.table(inventory)
```

output:

ProductID	ProductName	Quantity	Price(\$)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

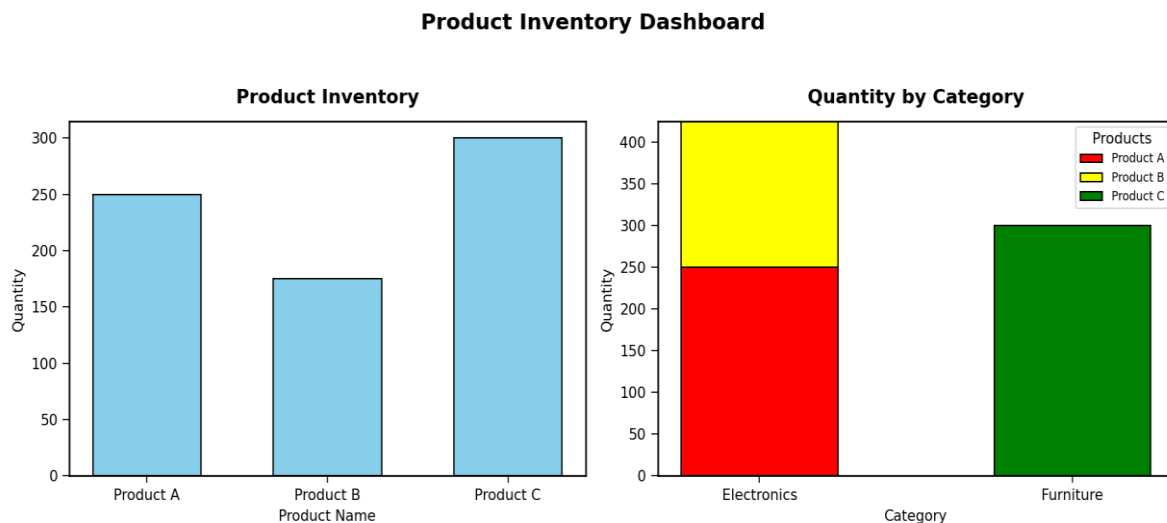
d) Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

R code (dashboard simulation):

```
par(mfrow = c(1, 2))
barplot(inventory$Quantity,
  names.arg = inventory$ProductName,
  col = "skyblue", xlab = "Product Name", ylab = "Quantity",
  main = "Product Inventory")
barplot(table_data,
  beside = FALSE, col = rainbow(ncol(table_data)),
  xlab = "Category", ylab = "Quantity",
```

```
main = "Quantity by Category")
legend("topright", legend = colnames(table_data), fill = rainbow(ncol(table_data)))
```

output:



Set 19: Survey Responses Analysis

a) Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

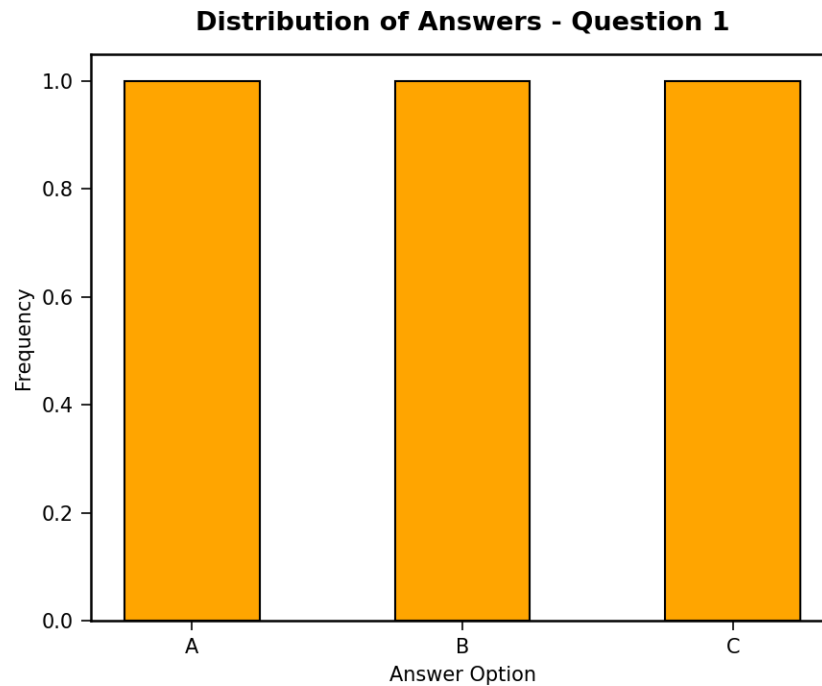
R code:

```
# Dataset
survey <- data.frame(
  SurveyID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

# Frequency of Question 1 answers
q1_count <- table(survey$Question1)

# Bar Chart
barplot(q1_count,
  col = "orange",
  xlab = "Answer Option",
  ylab = "Frequency",
  main = "Distribution of Answers - Question 1")
```

output:



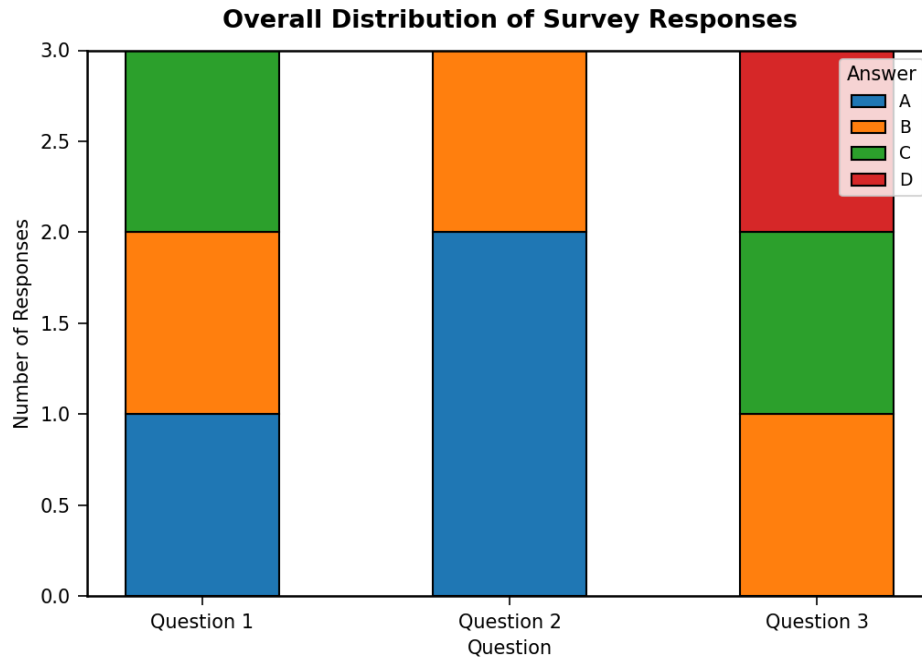
b) Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

R code:

```
# Combine counts for all questions
q_counts <- rbind(
  table(factor(survey$Question1, levels = c("A","B","C","D"))),
  table(factor(survey$Question2, levels = c("A","B","C","D"))),
  table(factor(survey$Question3, levels = c("A","B","C","D")))
)
rownames(q_counts) <- c("Question 1", "Question 2", "Question 3")

# Stacked Bar Chart
barplot(t(q_counts),
  beside = FALSE,
  col = rainbow(4),
  xlab = "Question",
  ylab = "Number of Responses",
  main = "Overall Distribution of Survey Responses")
legend("topright", legend = colnames(q_counts), fill = rainbow(4), title = "Answer")
```

output:



c) Build a table to show the survey response data for each survey. Label the table elements.

R code:

```
print(survey)
install.packages("gridExtra")
library(gridExtra)
grid.table(survey)
```

output:

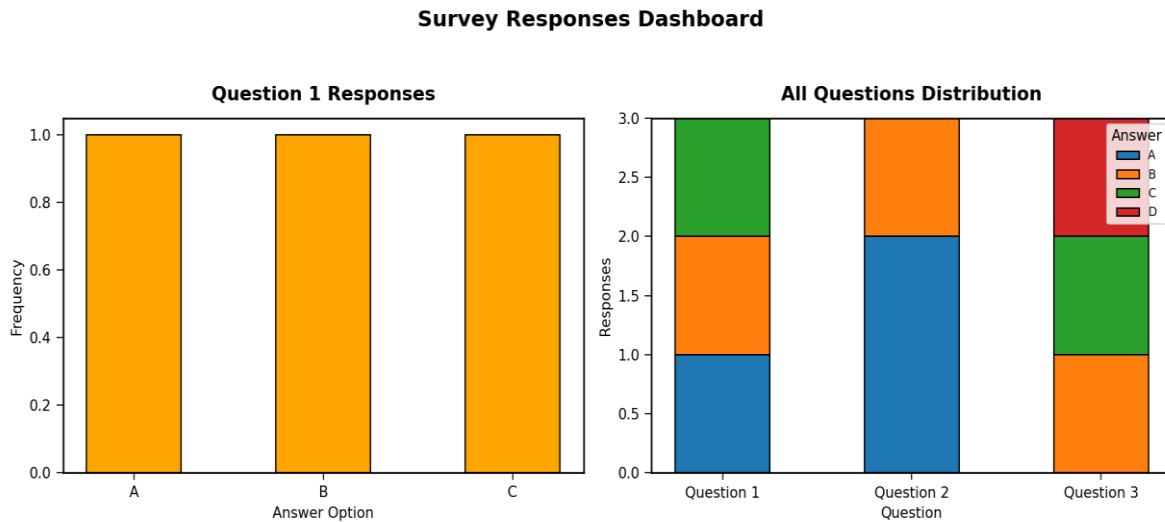
SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

d) Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

R code (dashboard simulation):

```
par(mfrow = c(1, 2))
barplot(q1_count,
  col = "orange", xlab = "Answer Option", ylab = "Frequency",
  main = "Question 1 Responses")
barplot(t(q_counts),
  beside = FALSE, col = rainbow(4),
  xlab = "Question", ylab = "Responses",
  main = "All Questions Distribution")
legend("topright", legend = colnames(q_counts), fill = rainbow(4))
```

output:



Set 20: Stock Analysis

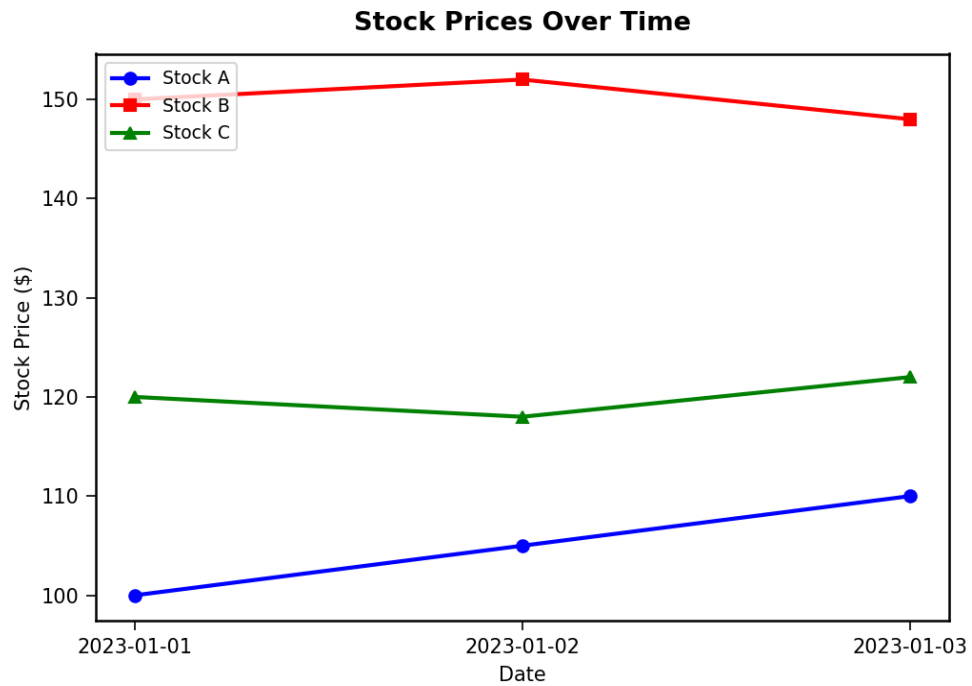
a) Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

R code:

```
# Dataset
stocks <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  StockA = c(100, 105, 110),
  StockB = c(150, 152, 148),
  StockC = c(120, 118, 122)
)

# Line Chart
plot(stocks$Date, stocks$StockA,
  type = "o", col = "blue", lwd = 2,
  ylim = range(c(stocks$StockA, stocks$StockB, stocks$StockC)),
  xlab = "Date", ylab = "Stock Price ($)",
  main = "Stock Prices Over Time")
lines(stocks$Date, stocks$StockB, type = "o", col = "red", lwd = 2)
lines(stocks$Date, stocks$StockC, type = "o", col = "green", lwd = 2)
legend("topleft", legend = c("Stock A", "Stock B", "Stock C"),
  col = c("blue", "red", "green"), lty = 1, pch = 1)
```

output:



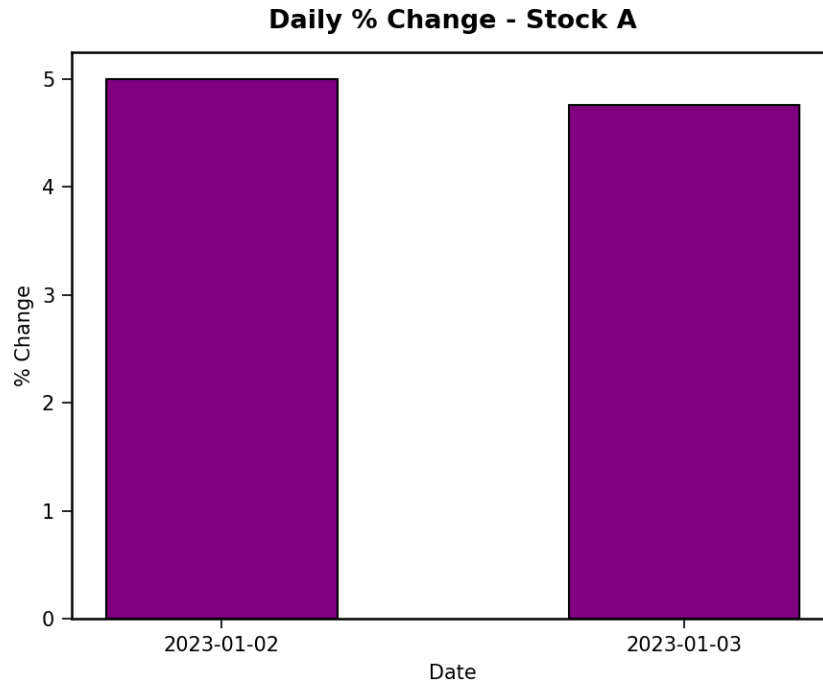
b) Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

R code:

```
# Daily % change for Stock A
pct_change <- c(NA, round(diff(stocks$StockA) / head(stocks$StockA, -1) * 100, 2))

# Bar Chart
barplot(pct_change[-1],
  names.arg = as.character(stocks$Date[-1]),
  col = "purple",
  xlab = "Date",
  ylab = "% Change",
  main = "Daily % Change - Stock A")
```

output:



c) Build a table to display the stock price data for each company over the given period. Label the table elements.

R code:

```
print(stocks)
install.packages("gridExtra")
library(gridExtra)
grid.table(stocks)
```

output:

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

d) Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

R code (dashboard simulation):

```
par(mfrow = c(1, 2))
plot(stocks$Date, stocks$StockA,
     type = "o", col = "blue", lwd = 2,
     ylim = range(c(stocks$StockA, stocks$StockB, stocks$StockC)),
     xlab = "Date", ylab = "Price ($)", main = "Stock Prices")
lines(stocks$Date, stocks$StockB, type = "o", col = "red", lwd = 2)
lines(stocks$Date, stocks$StockC, type = "o", col = "green", lwd = 2)
```

```

barplot(pct_change[-1],
  names.arg = as.character(stocks$Date[-1]),
  col = "purple", xlab = "Date", ylab = "% Change",
  main = "Stock A % Change")

```

output:

